THREE EQUIVALENT FORMULATIONS OF UNIVERSAL CLOSEDNESS FOR A QUASI-COMPACT SEPARATED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has three enumerated equivalent conditions including the displayed diagram of condition (3); the proof is a two-step deduction from the two cited lemmas.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- For separated morphisms the two formulations of the existence part agree. Lemma lemma-usual-enough (8615-8634), used at proof lines 8950-8952. Gives the equivalence of (2) and (3).
- The existence part of the criterion for quasi-compact quasi-separated morphisms. Lemma lemma-characterize-universally-closed-quasi-separated (8861-8872), used at proof lines 8953-8954. Gives the equivalence of (3) and (1).
- The three formulations of the valuative criterion. Definition definition-valuative-criterion and Schemes, Definition schemes-definition-valuative-criterion, used at proof lines 8934-8946. Fix the conditions being compared.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0A3X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12755–12755. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0A3W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12754–12754. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01KD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2046–2046. Navigation pointer into the

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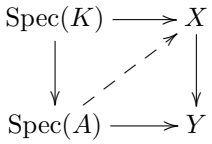
Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-compact and separated. Then the following are equivalent*

(1) *f is universally closed,*
(2) *the existence part of the valuative criterion holds as in Definition 03IX, and*
(3) *given any commutative solid diagram*



where A is a valuation ring with field of fractions K, there exists a dotted arrow, i.e., f satisfies the existence part of the valuative criterion as in Schemes, Definition 01KD.

Proof. Since f is separated parts (2) and (3) are equivalent by Lemma 0A3W. The equivalence of (3) and (1) follows from Lemma 0A3X. □